វិទ្យាស្ថាន សាយអិនស៍ អេន ថេកណាឡូជី អ៊ែតដ្វ៉ានស៍ ឌីវេឡុបម៉ិន
Institute of Science and Technology Advanced Development

Academic Year: 2025-2026
Spring - Microservice Architecture

COURSE DESCRIPTION

The Spring Microservices course will help you understand microservices architecture with event-driven architecture using Apache Kafka and how to build it using Spring Boot, Spring Cloud components, and Docker. By the end of this course, students will understand all the below topics:

COURSE CURRICULUM

N°	Content	Deadline	P	Q	H	T

1	**Spring Data MongoDB**
	- Understanding about MongoDB
	- SQL vs NoSQL
	- How to install MongoDB Server
	  - Manual Installation
	  - Docker Installation
	- Fundamental of Using MongoDB
	- How to use Spring Data MongoDB
	- Pagination & Sorting
	- Advanced Filtering
		25 Nov			✅	✅

2	**Build Reactive APIs with Spring WebFlux**
	- Overview of Reactive Programming
	  - What is Stream?
	  - What is Reactive Programming?
	  - Reactive Programming Origins
	  - Reactive Manifesto
	- Concurrency Model
	  - Block (sync) vs Non-Block (async)
	  - Mutable State
		27 Nov	✅		✅	✅

	o Thread Modeling - Backpressure o What is Backpressure? o Backpressure Strategies ▪ Buffer ▪ Drop ▪ Control - Reactive Stream API o Publisher o Subscriber o Subscription o Processor - Spring WebFlux o What is Spring WebFlux? o Applicability of Spring Web MVC vs Spring WebFlux o Reactive Stack vs Servlet Stack - Spring WebFlux Publishers o Mono Publisher o Flux Publisher o Publisher Operators - Spring WebFlux Programming Models o Annotation-based Reactive Components o Functional Routing and Handling - Server-Sent Event (SSE) o Understanding SSE o How to create an SSE element and consume it					
3	**OAuth 2 with Spring Ecosystem** ● Auth2 Overview ○ Principles of OAuth 2 ○ OAuth 2 Roles ○ OAuth 2 Scopes ○ OAuth 2 Token & Authorization Code ● Grant Types in OAuth 2 ○ Authorization Code Flow with PKCE ○ Client Credentials Flow ○ Refresh Token Flow ● How to implement OAuth 2 in Spring with Spring Data JPA and PostgreSQL ○ OAuth 2 Authorization Server ○ OAuth 2 Resource Server ○ OAuth 2 Client ● OAuth 2 Flow Testing with Postman Client Tool and NextJS	3 Dec	✅	✅	✅	✅

4	**Spring REST Clients**
	• **WebClient**
	  ○ What is WebClient?
	  ○ Benefits of WebClient
	  ○ WebClient Dependency
	• **HTTP Interface**
	  ○ What is an HTTP Interface?
	  ○ Why HTTP Interface?
	  ○ Setup with **<u>WebClient</u>**
	  ○ Exchange Methods
	  ■ @HttpExchange
	  ■ @GetExchange
	  ■ @PostExchange
	  ■ @PutExchange
	  ■ @PatchExchange
	  ■ @DeleteExchange
	  ○ Method Parameters
	  ■ @RequestHeader
	  ■ @PathVariable
	  ■ @RequestBody
	  ■ @RequestParam
	  ○ Setting up HTTP Interface Client
	  ○ How to make API calls with HTTP Interface
	  ○ How to handle the exception
	  ○ Multipart
		9 Dec			✅	✅

5	**Event-Driven with Apache Kafka** • Understanding about Event-Driven Architecture (EDA) • Introduction to Apache Kafka • Apache Kafka Use-Cases • Apache Kafka Installation Manually • Quick Start with Event and Topic • Docker Compose with Single and Cluster of Apache Kafka • Spring for Apache Kafka	12 Dec	✅	✅	✅	✅

6	**Spring Boot with Debezium**
	• Introduction to Debezium
	• Why need Debezium
	• Deploying the Debezium Cluster
	• Capturing data changes:
	  ○ From Oracle to PostgreSQL
	  ○ From CSV to Postgres
		18 Dec	✅		✅	

	○ From PostgreSQL to MongoDB ○ From PostgreSQL to ElasticSearch					

7	**Microservices Architecture (Part 1)**
	- The Evolutionary of Software Architectures
	  - What is software architecture?
	  - Why need software architecture?
	  - Evolutionary of software architecture
	- Software Development Business Models
	  - Product Strategy Model
	  - Revenue Model
	  - Distribution Model
	  - Service & Implementation Model
	- Cloud Service Models
	  - Software as a Service (SaaS)
	  - Platform as a Service (Paas)
	  - Infrastructure as a Service (Iaas)
	- Monolithic Architecture
	  - Overview
	  - Benefits
	  - Drawbacks
	- Microservice Architecture
	  - Overview
	  - Benefits
	  - Drawbacks
	  - Monolithic vs Microservices
	  - Key Differences
	  - Use-Cases
	- Microservice Architecture Patterns
	  - Decomposition
	  - Decompose by business capability
	  - Decompose by subdomain
	  - Service per team
	  - Database or Data Management
	  - Database per Service
	  - Shared database
	  - Saga
	  - API Composition
	  - CQRS
	  - Domain Event
	  - Event Sourcing
	  - Communication Style
	  - Remote Procedure Invocation
	  - Messaging
	  - Deployment Patterns
	  - Multiple service instances per host
	  - Service instance per host
	  - Service instance per VM
	  - Service instance per container
	  - Testing
		23 Dec		✅		✅

	▪ Service Component Test ▪ Automation Test o Cross-Cutting Concerns ▪ Microservice chassis ▪ Externalized configuration o Service Discovery ▪ Client-side discovery ▪ Server-side discovery ▪ Service registry o Reliability ▪ Circuit Breaker o Observability ▪ Log aggregation ▪ Application metric ▪ Audit logging ▪ Distributed tracing ▪ Exception tracking ▪ Health check API ▪ Log deployment and changes o API Gateway ▪ API Gateway ▪ Backend For Frontend (BFF) o UI Patterns ▪ Server-side page fragment composition ▪ Client-side UI component					
8	**Microservices Architecture (Part 2)** **Configuration Management** ɑ Introduction to Configuration Management ɑ Spring Cloud Config Overview o Config Server Overview ▪ Reading config from a file system ▪ Reading config from a GitHub repository ▪ Encryption & Description properties inside Config server ▪ Spring Vault Integration o Config Client Overview ɑ Refresh config at runtime using Spring Cloud Bus and Spring Cloud Config Monitor	30 Dec			✅	✅
9	**Microservices Architecture (Part 3)** **Service Discovery & Service Registration** ɑ Introduction to Microservices traffic ɑ Introduction to Service Discovery & Registration inside Microservices ɑ Understanding keywords in MS o Service Discovery & Registration	5 Jan			✅	✅

	o Client Side Service Discovery o Load-Balancing - Service Discovery Agent using Spring Eureka					
10	**Microservices Architecture (Part 4)** **API Gateway or Edge Server** - API Gateway Overview o What is an API Gateway? o Why use an API Gateway? o API Gateway Patterns - Introduction to Spring Cloud Gateway o Internal Architecture o Build API Gateway using Spring Cloud Gateway with Microservices - Service names with lowercase - Custom Routing - Response Header filter - Implement OAuth2 inside the Gateway Server - Implement BFF pattern with NextJS	13 Jan	✅		✅	
11	**Microservices Architecture (Part 5)** **Resilient / Reliability** - Introduction to the need for Resiliency inside Microservices - Typical use case or scenario for the need for Resiliency - Deep-dive on Circuit Breaker pattern - Three states of Circuit Breaker pattern - Implement Circuit Breaker pattern in Gateway - Implement Circuit Breaker pattern with Service Communication - Retry pattern in Gateway/Microservices - Rate Limiter pattern in Gateway/Microservices - Bulkhead pattern	19 Jan		✅	✅	✅
12	**Microservices Architecture (Part 6)** **Observability and Monitoring** - Introduction to Observability And Monitoring Of Microservices - Observability vs. Monitoring - Introduction to centralized logging or Log Aggregation in Microservices - Introduction to log management with Grafana, Loki, and Promtail - Implementing logging using Grafana, Loki & Promtail - Managing metrics & monitoring with Actuator, Micrometer, Prometheus & Grafana	26 Jan	✅		✅	✅

	ɑ Setup of micrometer inside microservices ɑ Setup of Prometheus inside microservices ɑ Prometheus & Grafana integration ɑ Grafana inbuilt & Custom Dashboards ɑ Distributed Tracing in microservices ɑ Introduction to OpenTelemetry ɑ Implement OpenTelemetry changes inside Microservices ɑ Implementing Tracing using Grafana, Tempo & OpenTelemetry					
13	**Microservices Architecture (Part 7)** **Event-driven microservices** ɑ Event-Driven Patterns and its implementation o Event-Sourcing o Aggregate DDD o SAGA o Command Query Responsibility Segregation (CQRS) o Change Data Capture (CDC)	29 Jan - 27 Feb	✅		✅	✅
14	**Microservices Architecture (Part 8)** **Deploying MS with Docker Compose** Deploy every part of Microservices with **docker-compose** with different environments such as dev, qa, prod, etc.	N/A	✅		✅	✅
15	**Case Study** **Core and Commercial Banking** Define your business idea in Core and Commercial Banking Design your Microservices Architecture with Event-Driven Architecture Implement Cloud-Native with Spring Eco-system	N/A			✅	✅